

◁ ROOF OF THE WORLD

The vast extent of the Tibetan Plateau, with its fringing ranges of lofty, snow-capped Himalayan Mountains, can be seen from space.

called a mantle plume rises and melts its way through the crust. The giant eruption may have been a contributing factor toward the extinction of the dinosaurs—the main culprit being the meteor strike—and scientists once believed that they might have been killed off by a cooling of the world's climate caused by gases released by this eruption.

Mountains rising

Following its separation from Africa, India moved north at a rate of about 8 in (20 cm) a year, faster than any other known plate movement. As it rammed into the southern edge of Asia, it exerted a huge impact.

In collisions between plates, one plate is usually subducted beneath the other. But these two plates were so closely matched in density that neither would sink, so the crust between them crumpled up into the highest and largest mountain range the world

▷ SOURCE OF THE GANGES

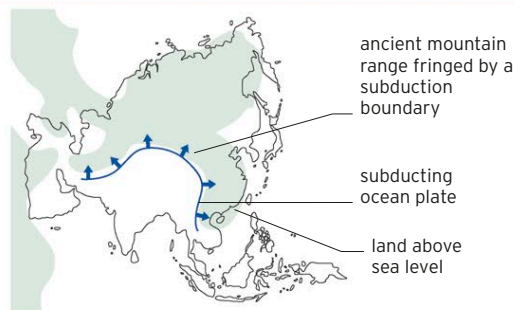
The rivers flowing down from the Himalayas, including the Ganges, sweep down 25 percent of all the world's sediment, to create wide, fertile plains.

has ever seen—the Himalayas. In just 50 million years, the rocks have been thrust up more than 6 miles (9 km) to create jagged peaks—including Everest, the world's highest mountain at 29,029 ft (8,848 m). In fact, the impact has not stopped the Indian Plate from moving—it still creeps forward at 1¼ in (3 cm) a year, and as a result the Himalayas rise ½ in (1 cm) every year. Geologists think the only reason they are not higher still is that, as it piles up, the rock also flows outward—in effect, the Himalayas are similar to the bow wave in front of a moving boat.

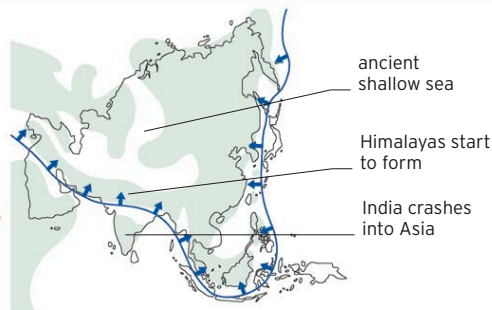
Violent volcanic eruptions in India 66.5 million years ago probably contributed to the extinction of the dinosaurs

BUILDING THE ASIAN CONTINENT

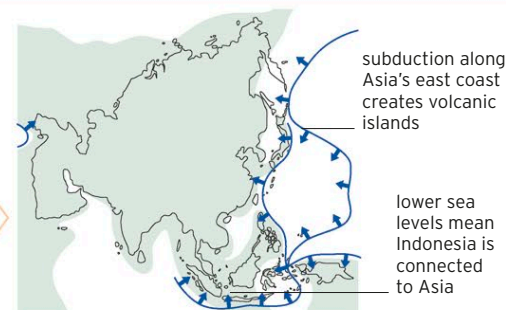
KEY — Convergent boundary



94 MILLION YEARS AGO Asia is much smaller than today. China has only recently joined, and India is still attached to Africa, far to the south. An ancient ocean plate to the south subducts beneath Asia.



50-40 MILLION YEARS AGO India crashes into the southern edge of Eurasia to create Asia's current shape and throw up the Himalayan Mountains. A shallow sea in central Asia recedes.



18,000 YEARS AGO The shallow seas that once separated Asia from Africa have receded completely. Lower sea levels due to water being locked away in ice sheets mean Indonesia is connected to Asia.